Data in Brief 39 (2021) 107672

Contents lists available at ScienceDirect

Data in Brief

journal homepage: www.elsevier.com/locate/dib

Data Article

Pillar 3–Pre-processed web server log file dataset of the banking institution

Michal Munk[a], Anna Pilkova[b], Ľubomír Benko[a,*], Petra Blazekova[b], Peter Svec[a]

[a] Constantine the Philosopher University in Nitra, Slovakia
[b] Comenius University in Bratislava, Slovakia

ARTICLE INFO

Article history:
Received 13 July 2021
Revised 29 November 2021
Accepted 30 November 2021
Available online 3 December 2021

Keywords:
Pillar 3
Basel II
Web usage mining
Data modeling
Multinomial Logit model

ABSTRACT

The dataset presented in this article represents the pre-processed web server log file of the commercial bank. The source of data is the web server of the bank and keeps access of web users starting the year 2009 till 2012. It contains accesses to the bank website during and after the financial crisis. Unnecessary data saved by the web server was removed to keep the focus only on the textual content of the website. Many variables were added to the original log file to make the analysis workable. To keep the privacy of website users, sensitive information in the log file were anonymized. The dataset offers a way to understand the behavior of stakeholders during and after the crisis and how they comply with the Basel regulations. The behavior of users can be modeled using the multinomial logit model, which is in detail described in the research article [1] related to this data article.

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DOI of original article: 10.1016/j.eswa.2021.115503
* Corresponding author.
E-mail address: lbenko@ukf.sk (Ľ. Benko).

https://doi.org/10.1016/j.dib.2021.107672
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Specifications Table

Subject	Computer Science (General)
Specific subject area	Modeling user behavior based on accesses to bank website content mandatory since Basel II Pillar 3 regulations
Type of data	Pre-processed log file
How data were acquired	Data were acquired from the web servers log files. Log files are stored as text files on the webserver and were raw copied for analysis. Using a simple script, the data were inserted/converted into an SQL database. Each line in the log file represents one row in the database table. After that, data were exported to a common CSV text file using the SQL query for exporting data.
Data format	Pre-processed data (.csv)
Parameters for data collection	Unnecessary data from the log file (access to graphic files, style sheets, java scripts and search engines robots) was removed. We used the Linux bash script to filter out lines containing specified keywords, e.g. png, jpg for images, css for stylesheets, js for javascript. To filter out access of search engines robots we searched for specific keywords for robot identification (e.g. BingBot, Googlebot, AOL, Baiduspider etc.), specific IP addresses based on a public list of robots IP addresses and access to robots.txt.
Description of data collection	The web server log file is created automatically by the web server when a user visits the website, and then the log file is copied for pre-processing and further analysis.
Data source location	Data related to Pillar 3 was obtained from the web server log files of domestic significant commercial banks operating in Slovakia. The log files are usually generated by the web server application as the text files or database entries. The exact form is depended on the technology the bank is using, which is, however, irrelevant for data analysis. The analysed data were obtained based on a request to the web server system administrator and were given as the text file.
Data accessibility	Munk, Michal; Pilkova, Anna; Benko, Ľubomír; Blazekova, Petra; Svec, Peter (2021), "Pillar 3: Pre-processed web server log file dataset of the banking institution", Mendeley Data, V1, doi:10.17632/5bvkm76sdc.1
Related research article	M. Munk, A. Pilkova, Ľ. Benko, P. Blažeková, P. Svec, Web usage analysis of Pillar 3 disclosed information by deposit customers in turbulent times, Expert Systems with Applications 185 (2021) 115503. doi:10.1016/j.eswa.2021.115503

Value of the Data

- The dataset contains data of web server log file of significant domestic commercial bank operating in Slovakia during the financial crisis and after the crisis and provides an option to analyse the stakeholders' behavior according to EU regulations.
- Researchers focused on the analysis and modeling of user behavior in the online environment can benefit from this data where the focus is on the time with examining the behavior of users in time.
- A reusable dataset for modeling the probabilities of the accesses to the web portal of the bank depending on time using the multinomial logit model which is a part of generalized linear models can be used for further insights and development of experiments.
- Dataset compromise Pillar 3 disclosure requirements quarterly and semi-annually; Pillar 3 related information; other information and describe market discipline categories regarding Basel II Pillar 3 regulations.
- Data pre-processing methods applied to this dataset can be used also in the field of educational data mining and learning analytics.

1. Data Description

The dataset is pre-processed standard web server log files. It contains only useful information for the analysis of user behavior. In the dataset, there are identified user sessions to distinct

visits of stakeholders. Pairs of visited part of the website and its referrer offer the way to restore the sitemap of the portal. As the source of the dataset in the web server of the commercial bank, the sensitive information is anonymized, but the researcher is still able to understand the meaning of this sensitive information. The dataset contains variables connected to the Basel II, Pillar 3 regulations. The Pillar 3 are specific regulatory disclosures requirements set out since the Basel II framework and incorporated into EU law and subsequent laws of the member states. Those regulations order the bank to publish various information and stakeholders can better understand the risk. The multinomial logit model offers the possibility to model bank website visitor behavior.

2. Experimental Design, Materials, and Methods

The research (in detail in Expert Systems with Applications [1]) is focused on modeling the bank visitors' behavior using a multinomial logit model [2]. The web usage analysis was done on log files obtained from the web servers of a bank institution. The log files consisted of a sample of 2 071 235 logged accesses that were obtained after data preparation. The analysis was oriented on examining the behavior of visitors over an extended period of time (2009-2012). The applied methodology to obtain and pre-process the web server log file is based on [3]. After obtaining log files from multiple servers, the data preparation was done in multiple steps.

The first step is **data cleaning**. All client requests on multimedia files, cascade styles sheet, java scripts and other non-content file types were removed using basic Linux commands. We are using a simple Linux bash script based on the grep command to filter our lines containing unnecessary keywords, e.g. png or jpg for images, js for JavaScript, css for stylesheets etc. All those requests form the web page design and are not valuable for the content analysis. The access of search engines robots was removed, too. Search engines robots are identified based on the *userAgent* or special access to robots.txt file on the webserver. We used the Linux bash script based on grep again, but we searched for robot identification, e.g. Google robots are identified as GoogleBot in the *userAgent* field, Microsoft robot is identified as Bingbot, etc. Some robots do not identify themselves but access directly to robots.txt. We search for access to robots.txt and marked down the IP of the origin to clean out all lines with this IP. We also used the public list of known IPs of search engines robots to filter them out. After the data cleaning phase, the raw data file only with accesses to html or pdf files of the portal is available. The associated file contains an already cleaned log. The algorithm for filtering data is as follows:

Foreach record from the logfile {
Parse record;
Foreach keyword from keywords filter_out (record, keyword);
Foreach IP from IPS filter_out(record, IP);
}

The second step in data preparation is **user/session identification**. To identify which lines of the log file, belong to the same user (visitor) and thus form a session it is needed to add fields into the log file. It was also needed to add a record to the log file. Some lines are missing because the user hits the back button in the browser. In this case, there is no request to the web server and the browser shows the content from its cache. To add missing lines the Path completion [4] was used. Visitors can be distinguished based on their IP address and the type of browser. While various users can share the same IP address or computer, the Reference Length method [5–8] for user/session was used. The published dataset does not contain a full IP address but only the first two bytes of the IP address. To keep the privacy of users according to GDPR the *anonIP* variable was created. The *anonIP* is a salted hash of the full IP. Researchers can analyze the dataset based on this hash but are not able to reveal the original IP. The first two bytes *ipPart* are enough to distinguish access of bank employees and other users. The bank employees connected from the internal bank network use the private network 10.238.x.x. To mark those users the *internal* variable was created in the dataset.

Table 1
Extract of the pre-processed log file.

Frame	Agent	ipPart	anonIP	length
883526	37885	81.83	9ad5c1b20796f7215dd25f597367df6e8a1a834c	–2
1005099	4528	85.248	1026980e36670a17d16758a71b6921cc304c5edd	8
819284	3416	78.98	211d24985f0162c0f2d6586db18e6bf09eeb2ecd	–1
1005099	4528	85.248	1026980e36670a17d16758a71b6921cc304c5edd	–1
603493	23	213.151	19190cdb4218cd39c9505622b43a5fde8f88cdf5	–2
73781	3530	10.238	141fa2a4792f00d8f4049b9cdd50417ad3dece5f	4
73781	3530	10.238	141fa2a4792f00d8f4049b9cdd50417ad3dece5f	7
73781	3530	10.238	141fa2a4792f00d8f4049b9cdd50417ad3dece5f	33
519356	18346	212.5	fd6504078435cf655f76cdb81cb920cdad3ef8b8	–1
877098	376288	80.94	40be9376d4efbc23b7d06c7561164ed1d9d09832	99

The values of internal can be

- 0, if the user is located outside the bank internal network,
- 1, if the user is in the bank internal network,
- 2, if the user is from the network of IT company that maintain and develop the web site.

To use the Reference Length Method, the *anonIP, Agent,* and *unixTime* are needed. The Agent is referred to as the type of web browser and the *unixTime* represents the standard unix time in seconds starting on January 1, 1970. The result of the Reference Length Method is the session represented by the *Frame-Agent-anonIP and* the *Length.*

The *Length* represents the session length calculated using the Reference Length method and its value represents the time in seconds a user spent on the website except for

- -1, which represents the end of the session caused by overrun of the session time threshold,
- -2, which represents the end of the session caused by the change of IP address,
- 0 represents robot access (crawling the website was too quick).

Table 1 shows the *Agent* variable which is created based on the *User_agent*. We kept both variables in the dataset. The session is a combination of *Frame-Agent-anonIP*.

The third step is **variables determination** for the user behavior analysis. A bank expert was asked to categorize every webpage according to the terminology used in the bank environment. The result of this process is the 19 different parts of the web represented by the *webPart* field each attached to 6 different categories – the variable *category* (Pricing list, Reputation, Business Conditions, Pillar3 related, Pillar3 disclosure requirements, We support..). Pillar 3 related information are those which are contained in the bank annual reports, minutes from general assembly meetings, prospect of emitent, information about group and information for banks. The Pillar 3 Disclosure Requirements consists of information on the Bank (organizational chart, information about employees or bank activities), financial information (financial statement information, information on asset quality, information on liquidity); information on risk management (risk strategies, policies, credit risk management). The webpage taxonomy is depicted in Table 2 and the detailed frequency of accesses to the web categories can be seen in Figs. 1 and 2. The algorithm for assigning variables is as follows:

Foreach record from the logfile {
Parse record;
Get IP, Agent, Unixtime from record;
Calculate Internal Access from record;
Calculate Length from record;
Get category from URL;
If not exist category {
Get category from Referrer;
}
If exist category {

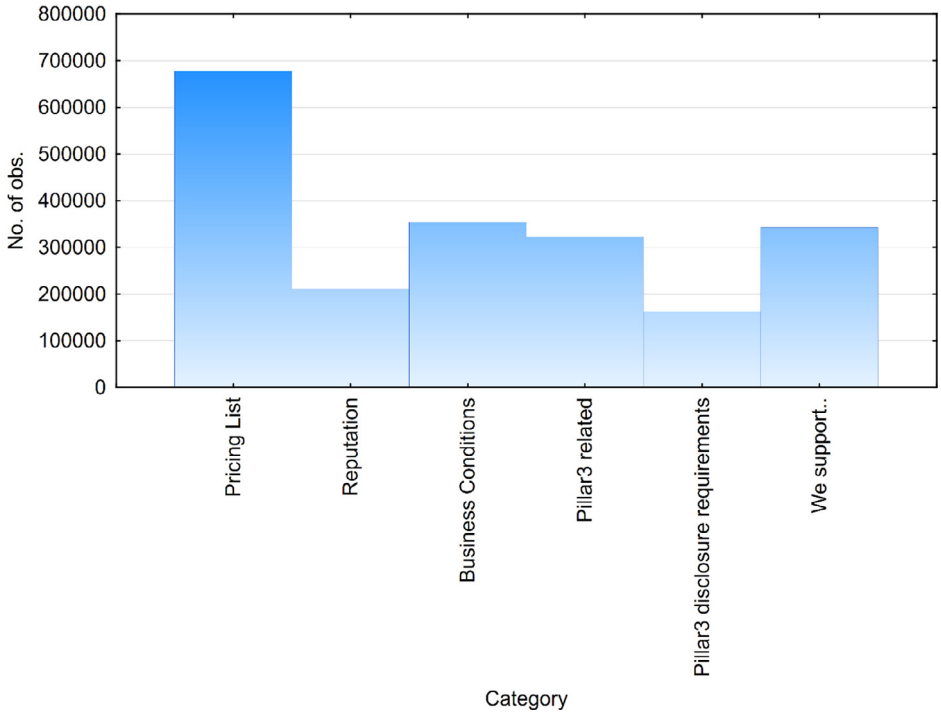


Fig. 1. Number of accesses to the examined web categories.

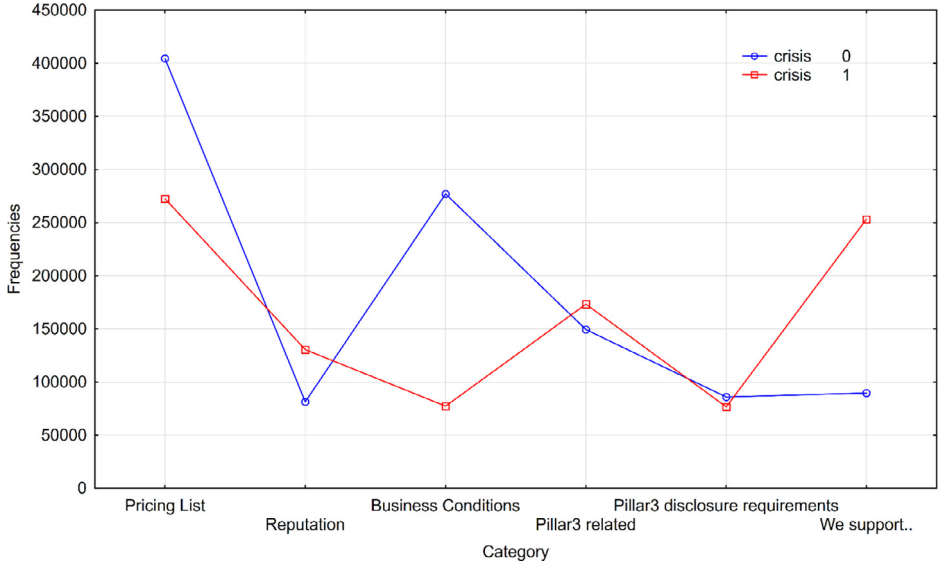


Fig. 2. Interaction plot for the accesses of the examined web categories.

Table 2
Taxonomy of the examined web portal content.

/Pillar 3 disclosure requirements/	/Pillar3 Q-terly Info/
/Pillar 3 disclosure requirements/	/Pillar3 Semi-annually Info/
/Pillar 3 related/	/Rating/
/Pillar 3 related/	/Annual reports/
/Pillar 3 related/	/Group/
/Pillar 3 related/	/Information for banks/
/Pillar 3 related/	/Emitent prospects/
/Pillar 3 related/	/General shareholder meeting/
/Pillar 3 related/	/Financial reports/
/Other/	

Assign webPart based on category;
Calculate Quartal from Timestamp;
}
}

The independent variables-predictors were identified as the next step. Field *Week* based on ISO 8601 with values of 0-53 was created. The value of 0 represents the case a week begins in the previous year. Another three fields were added to the dataset - *Quartal, Year* and *Year Quartal* which represents the number of quartal of the year.

To distinguish the time during the financial crisis and after the financial crisis, a field was added to the dataset. The *crisis* variable identifies the period of years during a financial crisis (the value is 1) and after a financial crisis (the value is 0).

To be able to restore the sitemap from the dataset, the URL and Referrer were kept. To keep privacy, only the salted hash of URL and the salted hash of Referrer is published. The same URL has the same hash *(urlHash)* and the same Referrer has the same hash *(referrerHash).* As the hash hides the extension of the file and it is not possible to distinguish html or pdf files or just the directory part of the URL the type of content was added before the anonymization took place. The fields *urlExt* and *referrefExt* were added to the dataset.

The obtained data file after the data preparation was used for web usage analysis. The investigated categorical dependent variable category that represents a group of web parts that deal with a similar issue was used. The analysis was oriented on examining the behavior of visitors over an extended period of time (2009-2012). The years 2009-2010 represent the years of the financial crisis. On the other hand, the years 2011-2012 represent the years after the financial crisis. The time independent variable was chosen as the variable week that was created from the date of the access of the visitor. The whole methodology of the experiment is described in detail in the MethodsX article [9]. Similar data pre-processing methods can be used also in other fields such as educational data mining or learning analytics where you work with data stored in log files from virtual learning environments [10,11]

Ethics Statement

The authors declare that the data presented in this article did not involve any use of human subjects, animal experiments nor data collected from social media platforms.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships which have, or could be perceived to have, influenced the work reported in this article.

CRediT Author Statement

Michal Munk: Conceptualization, Methodology, Formal analysis, Writing – review & editing; **Anna Pilkova:** Conceptualization, Writing – review & editing, Supervision; **Ľubomír Benko:** Methodology, Investigation, Writing – original draft; **Petra Blazekova:** Conceptualization, Writing – original draft; **Peter Svec:** Data curation, Resources, Writing – original draft.

Acknowledgments

This work was supported by the Scientific Grant Agency of the Ministry of Education of the Slovak Republic and of Slovak Academy of Sciences under contract VEGA-1/0776/18 and VEGA-1/0821/21.

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